

Worksheet 5

1. When should you use linear search vs. binary search? What do arrays need to have to use binary search?
2. `int[] arr = {12, 9, 5, 14, 8, 2, 13, 1}`
Sort the array with Selection, Insertion, and Merge Sort.

a. Selection Sort:

Pass 0 (original array):

Pass 1:

Pass 2:

Pass 3:

Pass 4:

Pass 5:

b. Insertion Sort:

Pass 0 (original array):

Pass 1:

Pass 2:

Pass 3:

Pass 4:

Pass 5:

c. Merge Sort:

Pass 0 (original array):

Pass 1:

Pass 2:

Pass 3:

Pass 4:

Pass 5:

3. `int[][] arr = {3, 6, 9, 13, 20}`

Find the value 20 with Binary Search.

4. `int[][] arr = {34, 56, 22, 13, 7, 45}`

Sort the array with Selection, Insertion, and Merge Sort.

a. Selection Sort:

Pass 0 (original array):

Pass 1:

Pass 2:

Pass 3:

Pass 4:

Pass 5:

b. Insertion Sort:

Pass 0 (original array):

Pass 1:

Pass 2:

Pass 3:

Pass 4:

Pass 5

c. Merge Sort:

Pass 0 (original array):

Pass 1:

Pass 2:

Pass 3:

Pass 4:

Pass 5:

